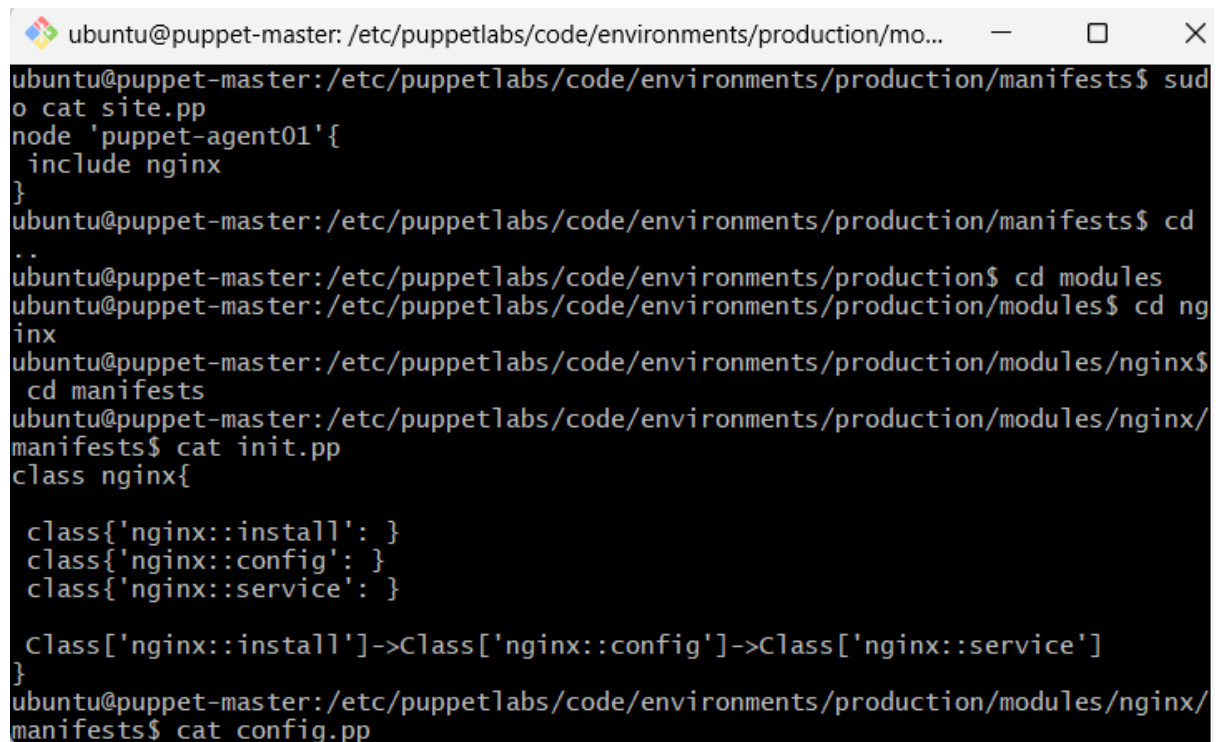


## Puppet Mastery Part-10: Extensive practice with Scenarios (Part-1)

### **Practical Scenario 1: Puppet-Managed Nginx Webpage Deployment**

**Scenario:** Use Puppet to install and manage Nginx on an agent, deploy an index.html webpage using an EPP template, and dynamically display the agent's **hostname**, **OS**, and **IP address** using Puppet facts.

**Master:**

A terminal window titled 'ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...' with standard window controls. The terminal shows a series of commands and their outputs. The user starts in the directory '/etc/puppetlabs/code/environments/production/manifests' and creates a file 'site.pp' using 'sudo cat site.pp'. The file contains a node 'puppet-agent01' that includes the 'nginx' module. The user then navigates through the directory structure: from 'manifests' to '..' (production), then to 'modules', then to 'nginx', and finally to 'manifests' again. In this final 'manifests' directory, they create 'init.pp' with 'cat init.pp'. The content of 'init.pp' defines a 'class nginx' with three subclasses: 'nginx::install', 'nginx::config', and 'nginx::service', each with an empty curly brace. A 'Class' line links them in sequence: 'Class['nginx::install']->Class['nginx::config']->Class['nginx::service']'. Finally, the user creates 'config.pp' with 'cat config.pp'.

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/manifests$ sudo
o cat site.pp
node 'puppet-agent01'{
  include nginx
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/manifests$ cd
..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production$ cd modules
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules$ cd ng
inx
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/nginx$
cd manifests
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/nginx/
manifests$ cat init.pp
class nginx{

  class{'nginx::install': }
  class{'nginx::config': }
  class{'nginx::service': }

  Class['nginx::install']->Class['nginx::config']->Class['nginx::service']
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/nginx/
manifests$ cat config.pp
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/nginx/manifests$ cat config.pp
class nginx::config{

  file{'/etc/nginx/sites-enabled/default':
    ensure=>absent,
    notify=>Service['nginx']
  }

  file{'/var/www/myapp':
    ensure=>directory,
    mode=>'0755',
    owner=>'root',
    group=>'root'
  }

  file{'/var/www/myapp/index.html':
    ensure=>file,
    mode=>'0644',
    owner=>'root',
    group=>'root',
    content=>epp('nginx/index.html.epp'),
    require=>File['/var/www/myapp']
  }
}
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/nginx/manifests$ cat install.pp
class nginx::install{

  file{'/etc/nginx/sites-available/myapp':
    ensure=>file,
    mode=>'0644',
    owner=>'root',
    group=>'root',
    content=>epp('nginx/myapp.conf.epp'),
    notify=>Service['nginx']
  }

  file{'/etc/nginx/sites-enabled/myapp':
    ensure=>link,
    target=>' /etc/nginx/sites-available/myapp',
    require=>File['/etc/nginx/sites-available/myapp']
  }
}

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/nginx/manifests$ cat install.pp
class nginx::install{

  package{'nginx':
    ensure=>installed
  }
}
```

```

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/nginx/manifests$ cat service.pp
class nginx::service{

    service{'nginx':
        ensure=>running,
        enable=>true
    }
}
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/nginx/manifests$ cd ..
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/nginx$ cd templates
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/nginx/templates$ cat myapp.conf.epp
server{
    listen 80;
    server_name _;
    root /var/www/myapp;
    index index.html;
}
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/nginx/templates$ ls
index.html.epp  myapp.conf.epp

```

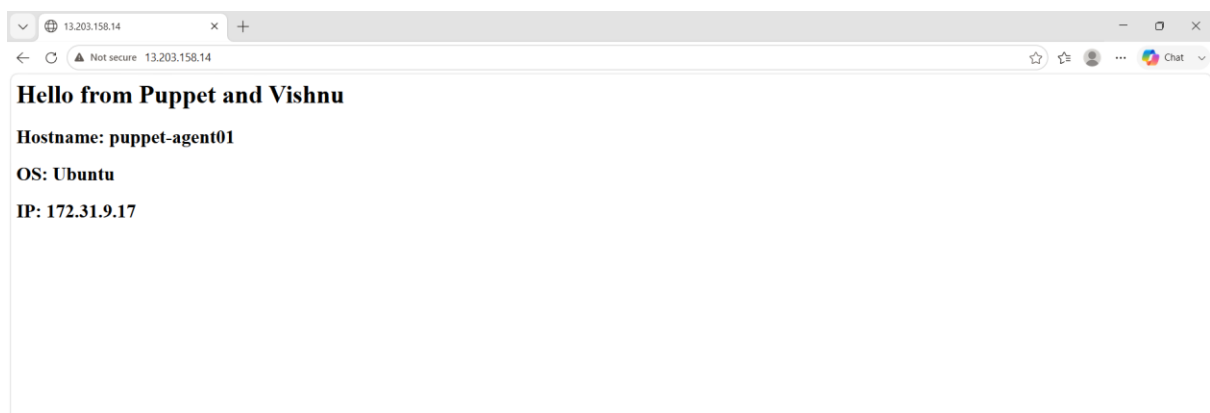
```

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/nginx/manifests$ cd ..
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/nginx$ cd templates
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/nginx/templates$ cat myapp.conf.epp
server{
    listen 80;
    server_name _;
    root /var/www/myapp;
    index index.html;
}
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/nginx/templates$ ls
index.html.epp  myapp.conf.epp
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/nginx/templates$ cat index.html.epp
<h1>Hello from Puppet and Vishnu</h1>
<h2>Hostname: <%= $facts['networking']['hostname'] %></h2>
<h2>OS: <%= $facts['os']['name'] %></h2>
<h2>IP: <%= $facts['networking']['ip'] %></h2>
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/nginx/templates$

```

## Agent-01:

```
ubuntu@puppet-agent01: ~  
ubuntu@puppet-agent01:~$ sudo /opt/puppetlabs/bin/puppet agent -t  
Info: Using environment 'production'  
Info: Retrieving pluginfacts  
Info: Retrieving plugin  
Notice: Requesting catalog from puppet-master:8140 (172.31.2.86)  
Notice: Catalog compiled by puppet-master  
Info: Caching catalog for puppet-agent01  
Info: Applying configuration version '1787153021'  
Notice: /Stage[main]/Nginx::Config/File[/var/www/myapp/index.html]/content:  
--- /var/www/myapp/index.html 2026-08-19 15:11:37.135271379 +0000  
+++ /tmp/puppet-file20260819-3468-7r8df4 2026-08-19 15:23:43.017357592 +0000  
@@ -1,4 +1,4 @@  
<h1>Hello from Puppet and Vishnu</h1>  
<h2>Hostname: puppet-agent01</h2>  
<h2>OS: Ubuntu</h2>  
-<h2>IP: 172.31.9.17</h2>  
+<h2>IP: 172.31.9.17</h2>  
  
Notice: /Stage[main]/Nginx::Config/File[/var/www/myapp/index.html]/content: content changed '{sha256}83a8c399808f3d43aa02d1ac29deb5fff60093ffc896632a18f5b62cadf22bcb' to '{sha256}15f300c0aa0af0be4fcc209f48c36d69dd7bddb0a86083887c154dc7e4e17ae8'  
Notice: Applied catalog in 0.47 seconds
```



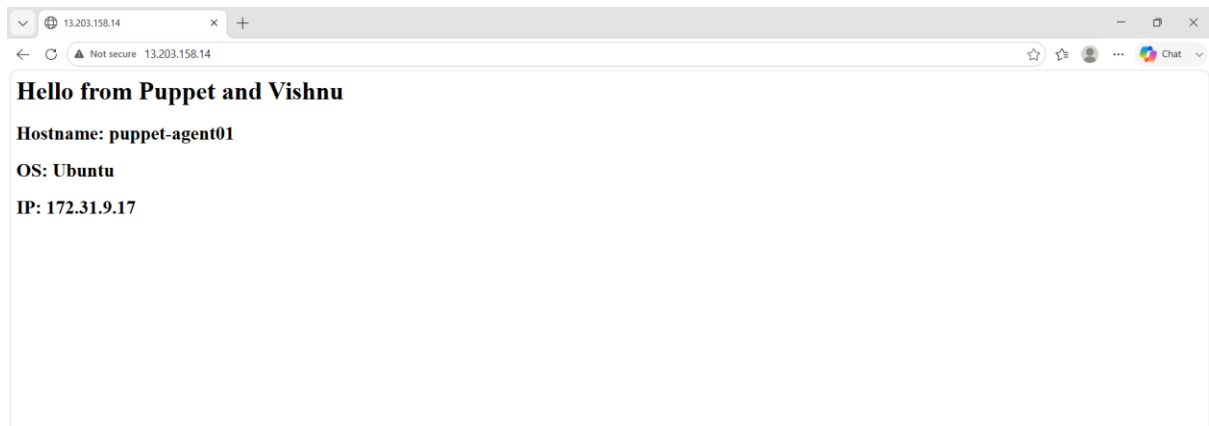
## 5-min practicals with Nginx and Apache:

1. Nginx (Note: since my previous practical was based on production level ideation the ensure absent on sites-available removed the default so we are just getting it back and creating the sym-links if the previous practical wasn't done the config.pp would be even simpler just the index.html.epp would be needed)

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...  
manifests$ sudo cat config.pp  
class nginx::config{  
  
  file{'/etc/nginx/sites-available':  
    ensure=>directory  
  }  
  
  file{'/etc/nginx/sites-available/default':  
    ensure=>file,  
    content=>epp('nginx/myapp.conf.epp'),  
    require=>File['/etc/nginx/sites-available']  
  }  
  
  file{'/etc/nginx/sites-enabled/default':  
    ensure=>link,  
    target=>'/etc/nginx/sites-available/default',  
    notify=>Service['nginx'],  
    require=>File['/etc/nginx/sites-available/default']  
  }  
  
  file{'/var/www/html/index.html':  
    ensure=>file,  
    owner=>'root',  
    group=>'root',
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...  
  
file{'/etc/nginx/sites-available/default':  
  ensure=>file,  
  content=>epp('nginx/myapp.conf.epp'),  
  require=>File['/etc/nginx/sites-available']  
}  
  
file{'/etc/nginx/sites-enabled/default':  
  ensure=>link,  
  target=>'/etc/nginx/sites-available/default',  
  notify=>Service['nginx'],  
  require=>File['/etc/nginx/sites-available/default']  
}  
  
file{'/var/www/html/index.html':  
  ensure=>file,  
  owner=>'root',  
  group=>'root',  
  mode=>'0644',  
  content=>epp('nginx/index.html.epp')  
}  
}  
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/nginx/  
manifests$ |
```

```
ubuntu@puppet-agent01: ~  
ubuntu@puppet-agent01:~$ sudo /opt/puppetlabs/bin/puppet agent -t  
Info: Using environment 'production'  
Info: Retrieving pluginfacts  
Info: Retrieving plugin  
Notice: Requesting catalog from puppet-master:8140 (172.31.2.86)  
Notice: Catalog compiled by puppet-master  
Info: Caching catalog for puppet-agent01  
Info: Applying configuration version '1787157536'  
Notice: /Stage[main]/Nginx::Config/File[/etc/nginx/sites-enabled/default]/target  
: target changed '/etc/nginx/sites-availble/default' to '/etc/nginx/sites-availa  
ble/default'  
Info: /Stage[main]/Nginx::Config/File[/etc/nginx/sites-enabled/default]: Schedul  
ing refresh of Service[nginx]  
Notice: /Stage[main]/Nginx::Service/Service[nginx]/ensure: ensure changed 'stopp  
ed' to 'running' (corrective)  
Info: /Stage[main]/Nginx::Service/Service[nginx]: Unscheduling refresh on Servic  
e[nginx]  
Notice: Applied catalog in 0.78 seconds  
ubuntu@puppet-agent01:~$ curl http://localhost  
<h1>Hello from Puppet and Vishnu</h1>  
<h2>Hostname: puppet-agent01</h2>  
<h2>OS: Ubuntu</h2>  
<h2>IP: 172.31.9.17</h2>  
ubuntu@puppet-agent01:~$ |
```



## 2. Apache

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/apache
/manifests$ cat init.pp
class apache{

  class{'apache::install': }
  class{'apache::config': }
  class{'apache::service': }

  Class['apache::install']->Class['apache::config']->Class['apache::service']
}
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/apache
/manifests$ cat config.pp
class apache::config{

  file{'/var/www/html/index.html':
    ensure=>file,
    owner=>'root',
    group=>'root',
    mode=>'0644',
    content=>epp('apache/index.html.epp')
  }
}
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/apache
/manifests$ cat service.pp
class apache::service{
```

```

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/apache/manifests$ cat service.pp
class apache::service{

    service{'apache2':
        ensure=>running,
        enable=>true
    }
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/apache/manifests$ cat install.pp
class apache::install{

    package{'apache2':
        ensure=>installed
    }
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/apache/manifests$ cd ..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/apache$ cd templates
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/apache/templates$ ls
index.html.epp  myapp.conf.epp

```

```

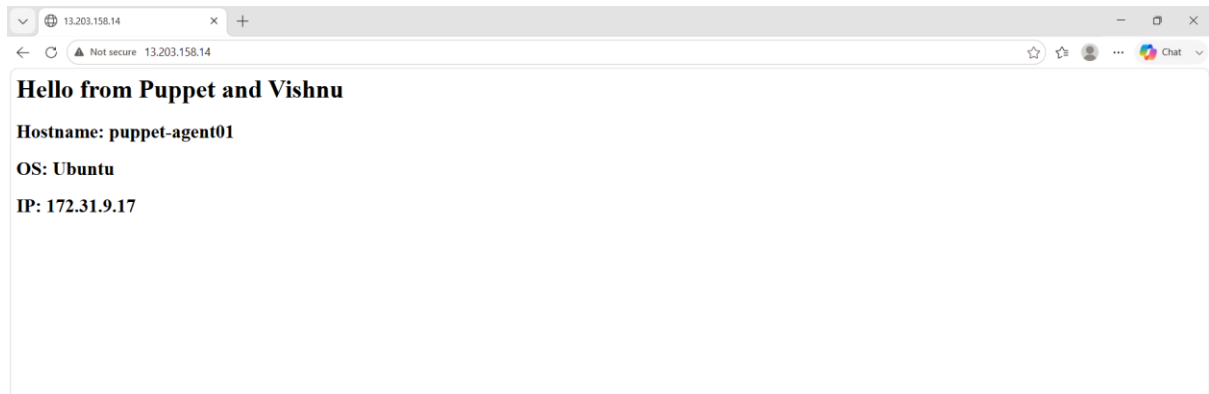
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/apache/manifests$ cat install.pp
class apache::install{

    package{'apache2':
        ensure=>installed
    }
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/apache/manifests$ cd ..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/apache$ cd templates
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/apache/templates$ ls
index.html.epp  myapp.conf.epp
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/apache/templates$ cat index.html.epp
<h1>Hello from Puppet and Vishnu</h1>
<h2>Hostname: <%= $facts['networking']['hostname'] %></h2>
<h2>OS: <%= $facts['os']['name']%></h2>
<h2>IP: <%= $facts['networking']['ip']%></h2>
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/apache/templates$

```

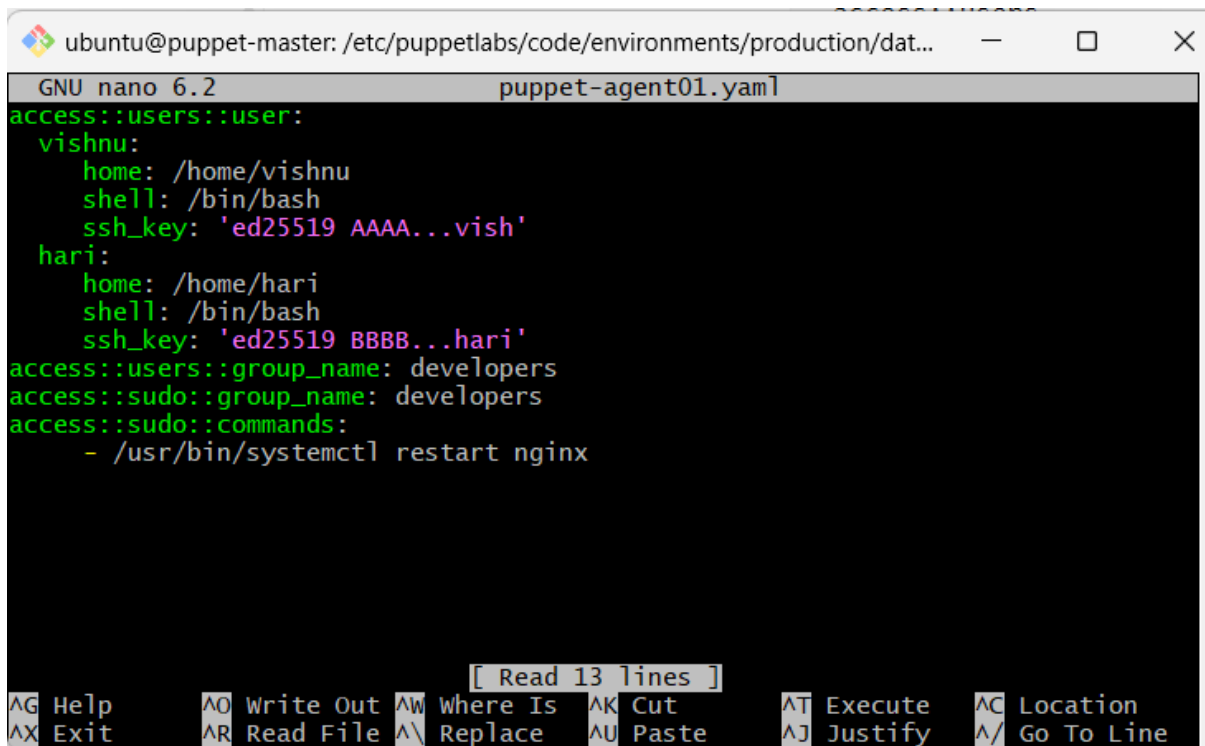


```
ubuntu@puppet-agent01:~$ sudo /opt/puppetlabs/bin/puppet agent -t
Info: Using environment 'production'
Info: Retrieving pluginfacts
Info: Retrieving plugin
Notice: Requesting catalog from puppet-master:8140 (172.31.2.86)
Notice: Catalog compiled by puppet-master
Info: Caching catalog for puppet-agent01
Info: Applying configuration version '1787158683'
Notice: /Stage[main]/Apache::Service/Service[apache2]/ensure: ensure changed 'stopped' to 'running'
Info: /Stage[main]/Apache::Service/Service[apache2]: Unscheduling refresh on Service[apache2]
Notice: Applied catalog in 0.78 seconds
ubuntu@puppet-agent01:~$ curl http://localhost
<h1>Hello from Puppet and Vishnu</h1>
<h2>Hostname: puppet-agent01</h2>
<h2>OS: Ubuntu</h2>
<h2>IP: 172.31.9.17</h2>
ubuntu@puppet-agent01:~$
```



## Scenario-3: Centralized User, Group, SSH Key & Sudo Management

**Master:**



```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/dat...
GNU nano 6.2 puppet-agent01.yaml
access::users::user:
  vishnu:
    home: /home/vishnu
    shell: /bin/bash
    ssh_key: 'ed25519 AAAA...vish'
  hari:
    home: /home/hari
    shell: /bin/bash
    ssh_key: 'ed25519 BBBB...hari'
access::users::group_name: developers
access::sudo::group_name: developers
access::sudo::commands:
  - /usr/bin/systemctl restart nginx

[ Read 13 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

```

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/dat...
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/manifests$ sudo cat site.pp
node 'puppet-agent01'{
  include access
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/manifests$ cd ..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production$ cd modules/access/manifests
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/access/manifests$ ls
init.pp  sudo.pp  users.pp
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/access/manifests$ sudo cat init.pp
class access{
  contain access::users
  contain access::sudo

  Class['access::users']->Class['access::sudo']
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/access/manifests$ sudo cat users.pp
class access::users(
  Hash $user,

```

```

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/dat...
class access::users(
  Hash $user,
  String $group_name='developers'
){
  group{$group_name:
    ensure=>present
  }

  $user.each |String $username,Hash $user_data| {
    user{$username:
      ensure=>present,
      gid=>$group_name,
      shell=>$user_data['shell'],
      home=>$user_data['home'],
      managehome=>true
    }

    file{"${user_data['home']}/.ssh":
      ensure=>directory,
      owner=>$username,
      group=>$group_name,
      mode=>'0700',
      require=>User[$username]
    }
  }
}

```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/dat...
file{"${user_data['home']}/.ssh/authorized_keys":
  ensure=>file,
  owner=>$username,
  group=>$group_name,
  mode=>'0600',
  content=>$user_data['ssh_key'],
  require=>File["${user_data['home']}/.ssh"]
}
}
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/access
/manifests$ sudo cat sudo.pp
class access::sudo(
  String $group_name='developers',
  Array[String] $commands=[]
){

  file{'/etc/sudoers.d/developers':
    ensure=>file,
    owner=>'root',
    group=>'root',
    mode=>'0440',
    content=>epp('access/sudoers.epp',
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/dat...
{
  'group_name'=>$group_name,
  'commands'=>$commands
}
),
validate_cmd=>'/usr/sbin/visudo -cf %'
}
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/access
/manifests$ cd ..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/access
$ cd templates
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/access
/templates$ sudo cat sudoers.epp
%<%= $group_name %> ALL=(ALL) <%= $commands[0] %>
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/access
/templates$ cd ..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/access
$ cd ..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules$ cd ..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production$ cd data
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/data$ ls
nodes
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/data$ cd nodes
```

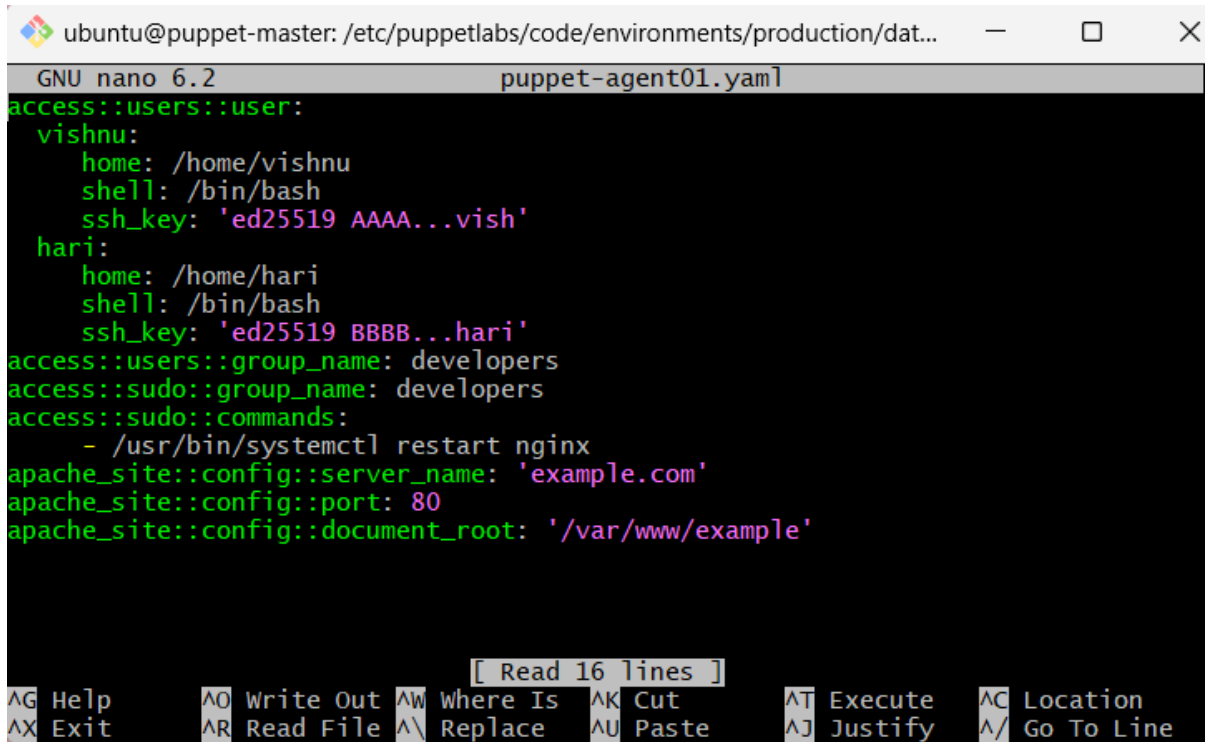
## Agent:

```
ubuntu@puppet-agent01: ~  
ubuntu@puppet-agent01:~$ sudo /opt/puppetlabs/bin/puppet agent -t  
Info: Using environment 'production'  
Info: Retrieving pluginfacts  
Info: Retrieving plugin  
Notice: Requesting catalog from puppet-master:8140 (172.31.2.86)  
Notice: Catalog compiled by puppet-master  
Info: Caching catalog for puppet-agent01  
Info: Applying configuration version '1787214776'  
Notice: /Stage[main]/Access::Users/Group[developers]/ensure: created  
Notice: /Stage[main]/Access::Users/User[vishnu]/ensure: created  
Notice: /Stage[main]/Access::Users/File[/home/vishnu/.ssh]/ensure: created  
Notice: /Stage[main]/Access::Users/File[/home/vishnu/.ssh/authorized_keys]/ensure: defined content as '{sha256}a1d33ae393e3022cb5b828b0e3d4872d8d0b79c5010dfd8d2ad49efa12456020'  
Notice: /Stage[main]/Access::Users/User[hari]/ensure: created  
Notice: /Stage[main]/Access::Users/File[/home/hari/.ssh]/ensure: created  
Notice: /Stage[main]/Access::Users/File[/home/hari/.ssh/authorized_keys]/ensure: defined content as '{sha256}8fa69515b629c40d2141414c208c15f66b5b090971726df7f6fe367278af8478'
```

```
ubuntu@puppet-agent01:~$ getent group developers  
developers:x:1001:  
ubuntu@puppet-agent01:~$ id vishnu  
uid=1001(vishnu) gid=1001(developers) groups=1001(developers)  
ubuntu@puppet-agent01:~$ id hari  
uid=1002(hari) gid=1001(developers) groups=1001(developers)  
ubuntu@puppet-agent01:~$ getent passwd vishnu  
vishnu:x:1001:1001::/home/vishnu:/bin/bash  
ubuntu@puppet-agent01:~$ getent passwd hari  
hari:x:1002:1001::/home/hari:/bin/bash  
ubuntu@puppet-agent01:~$ ls -ld /home/vishnu/.ssh  
ls: cannot access '/home/vishnu/.ssh': Permission denied  
ubuntu@puppet-agent01:~$ sudo ls -ld /home/vishnu/.ssh  
drwx----- 2 vishnu developers 4096 Aug 20 08:32 /home/vishnu/.ssh  
ubuntu@puppet-agent01:~$ sudo ls -l /home/vishnu/.ssh/authorized_keys  
-rw----- 1 vishnu developers 19 Aug 20 08:32 /home/vishnu/.ssh/authorized_keys  
ubuntu@puppet-agent01:~$ sudo cat /home/vishnu/.ssh/authorized_keys  
ed25519 AAAA...vishu  
ubuntu@puppet-agent01:~$ sudo cat /home/hari/.ssh/authorized_keys  
ed25519 BBBB...hari  
ubuntu@puppet-agent01:~$
```

## Scenario-4: Cross-Platform Apache Virtual Host & Website Deployment

**Master:**



```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/dat...
GNU nano 6.2 puppet-agent01.yaml
access::users::user:
  vishnu:
    home: /home/vishnu
    shell: /bin/bash
    ssh_key: 'ed25519 AAAA...vish'
  hari:
    home: /home/hari
    shell: /bin/bash
    ssh_key: 'ed25519 BBBB...hari'
access::users::group_name: developers
access::sudo::group_name: developers
access::sudo::commands:
  - /usr/bin/systemctl restart nginx
apache_site::config::server_name: 'example.com'
apache_site::config::port: 80
apache_site::config::document_root: '/var/www/example'
```

[ Read 16 lines ]

|                |                     |                    |                 |                   |                      |
|----------------|---------------------|--------------------|-----------------|-------------------|----------------------|
| <b>^G</b> Help | <b>^O</b> Write Out | <b>^W</b> Where Is | <b>^K</b> Cut   | <b>^T</b> Execute | <b>^C</b> Location   |
| <b>^X</b> Exit | <b>^R</b> Read File | <b>^_</b> Replace  | <b>^U</b> Paste | <b>^J</b> Justify | <b>^_</b> Go To Line |

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/apache_site/manifests$ sudo cat init.pp
class apache_site{

    contain apache_site::install
    contain apache_site::config
    contain apache_site::service

    Class['apache_site::install']->Class['apache_site::config']->
    Class['apache_site::service']
}

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/apache_site/manifests$ sudo cat config.pp
class apache_site::config(
    String $document_root,
    String $server_name,
    Integer $port
){

    file{$document_root:
        ensure=>directory,
        owner=>root,
        group=>root,
        mode=>'0755'
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/apache_site/manifests$ sudo cat config.pp
    file{"${document_root}/index.html":
        ensure=>file,
        owner=>root,
        group=>root,
        mode=>'0644',
        content=>epp('apache_site/index.html.epp',
        {
            'port'=>$port
        }),
        require=>File[$document_root]
    }

    $config_file = "/etc/apache2/sites-available/${server_name}.conf"

    file{$config_file:
        ensure=>file,
        owner=>'root',
        group=>'root',
        content=>epp('apache_site/vhost.conf.epp',
        {
            'port'=>$port,
            'document_root'=>$document_root,
            'server_name'=>$server_name
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...  
'document_root'=>$document_root,  
'server_name'=>$server_name  
}),  
require=>File[$document_root]  
}  
  
if $facts['os']['family'] == 'Debian' {  
  file{"/etc/apache2/sites-enabled/${server_name}.conf":  
    ensure=>link,  
    target=>config_file,  
    require=>File[$config_file]  
  }  
}  
}  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/apache  
_site/manifests$ sudo cat service.pp  
class apache_site::service{  
  
  $service = $facts['os']['family'] ? {  
    'Debian'=>'apache2',  
    'RedHat'=>'httpd',  
    default=>'apache2'  
  }  
}
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...  
  
service{$service:  
  ensure=>running,  
  enable=>true  
}  
}  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/apache  
_site/manifests$ sudo cat install.pp  
class apache_site::install{  
  
  $package = $facts['os']['family'] ? {  
    'Debian'=>'apache2',  
    'RedHat'=>'httpd',  
    default=>'apache2'  
  }  
  
  package{$package:  
    ensure=>installed  
  }  
}  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/apache  
_site/manifests$ cd ..  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/apache  
_site$ cd templates
```



```

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/apache_site/manifests$ cd ..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/apache_site$ cd templates
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/apache_site/templates$ ls
index.html.epp  vhost.conf.epp
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/apache_site/templates$ sudo cat index.html.epp
<h1>Vishnu's Apache production level config</h1>
<h2>Server serving on: <%= $port %></h2>
<h2>Thanks to Puppet</h2>
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/apache_site/templates$ sudo cat vhost.conf.epp
<VirtualHost *: <%= $port %>>
  ServerName <%= $server_name %>
  DocumentRoot <%= $document_root %>

  <Directory <%= $document_root %>>
    Require all granted
  </Directory>
</VirtualHost>
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/apache_site/templates$

```

## Agent:

```

ubuntu@puppet-agent01: ~
Info: Using environment 'production'
Info: Retrieving pluginfacts
Info: Retrieving plugin
Notice: Requesting catalog from puppet-master:8140 (172.31.2.86)
Notice: Catalog compiled by puppet-master
Info: Caching catalog for puppet-agent01
Info: Applying configuration version '1787227958'
Notice: /Stage[main]/Apache_site::Service/Service[apache2]/ensure: ensure changed 'stopped' to 'running'
Info: /Stage[main]/Apache_site::Service/Service[apache2]: Unscheduling refresh on Service[apache2]
Notice: Applied catalog in 0.77 seconds
ubuntu@puppet-agent01:~$ curl -H "Host: example.com" http://3.110.155.98\
>
<h1>Hello from Puppet and Vishnu</h1>
<h2>Hostname: puppet-agent01</h2>
<h2>OS: Ubuntu</h2>
<h2>IP: 172.31.9.17</h2>
ubuntu@puppet-agent01:~$ curl -H "Host: example.com" http://3.110.155.98
<h1>Hello from Puppet and Vishnu</h1>
<h2>Hostname: puppet-agent01</h2>
<h2>OS: Ubuntu</h2>
<h2>IP: 172.31.9.17</h2>
ubuntu@puppet-agent01:~$ |

```